Experiment 5: Set Up D3.js and Create Basic Visualizations

Aim:

To set up D3.js, load a dataset, and create basic visualizations such as bar charts and line charts using D3's data loading and selection functions.

Theory:

D3.js (Data-Driven Documents) is a powerful JavaScript library used to create dynamic and interactive data visualizations in web browsers. It allows developers to bind data with HTML, SVG, and CSS elements to visually represent information.

D3.js works on the concept of data binding, where data values are linked to visual elements such as bars, lines, or circles. When the data changes, the visualization updates automatically.

The library supports loading data from sources like CSV and JSON files and can be used to create charts such as bar charts, line charts, scatter plots, and pie charts. It mainly uses SVG (Scalable Vector Graphics) to display visual elements.

Thus, D3.js helps developers create custom and interactive visualizations that make data easier to explore and understand**.**

Procedure:

1. Install a code editor such as **Visual Studio Code**.
2. Create an **HTML file** and include the **D3.js library** using a CDN link.
3. Prepare a sample dataset in **CSV or JSON format**.

4. Use D3's **data loading functions** to import the dataset.
5. Create an **SVG container** to display the visualization.
6. Bind the dataset to visual elements using **D3 selections**.
7. Create a **bar chart** by mapping data values to rectangular bars.
8. Create a **line chart** to represent data trends.
9. Add axes, labels, and basic styling for better visualization.
10. Run the HTML file in a browser to view the visualizations.

Conclusion:

In this experiment, **D3.js** was successfully set up and used to load a dataset and create basic visualizations such as **bar charts and line charts**. Data binding and selection techniques were used to map data values to graphical elements.

This experiment demonstrated how D3.js enables the creation of **interactive and customizable visualizations** for web-based data analysis.